

Homework

Question 1

Think of a recent situation in which you changed your mind. Make a Bayesian knowledge-building diagram that includes your prior information, explain how new data helped you change your mind and your posterior conclusion.

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Question 2

There are several data scientist openings at a heavily hyped company. Having read the job description, you know for a fact that you are qualified for the position (this is your data). Your goal is to ascertain whether you will actually be offered a position (this is your hypothesis).

- a. From the perspective of someone using frequentist thinking, what question is answered in testing the hypothesis that you'll be offered the position?
- b. Repeat part (a) within a Bayesian framework
- c. Which question would you rather have the answer to: the frequentist or the Bayesian? Explain.

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Question 3

- a. Identify a topic that you know about (e.g., a sport, a school subject, music).
- b. Identify a hypothesis about this subject.
- c. How would your current expertise inform your conclusion about this hypothesis?
- d. Which framework are you employing here, Bayesian or frequentist?

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Question 4

A university health center is considering whether to recommend a costly new screening test for a rare disease among students. Historical data suggest that only 2% of students have the disease. The test correctly identifies diseased students 95% of the time. The test falsely identifies healthy students as diseased 8% of the time. A student tests positive.

- a. What is the probability the student actually has the disease?
- b. Should the health center automatically recommend treatment after one positive test?
- c. What are the risks of a false positive? a false negative?
- d. How would the decision change if the disease were more common?

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Question 5

Suppose two analysts disagree about the probability of a cyberattack before seeing evidence:

- Analyst A believes the probability is 1%.
- Analyst B believes the probability is 20%.

Then, both observe the same security alert.

- a. How can Bayesian reasoning explain why the analysts may still reach different conclusions?
- b. What role do priors play in decision-making under uncertainty?
- c. When is it reasonable for two people to maintain different posterior beliefs after observing the same data?
- d. How do costs and risks influence the final decision, beyond just the posterior probability?

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Question 6

Your friend just became interested in Bayesian statistics.

Explain the following to them:

- a. Why is Bayesian statistics useful?
- b. What are the similarities in Bayesian and frequentist statistics?